FACE RECOGNITION BASED SECURED ONLINE VOTING SYSTEM USING MACHINE LEARNING TECHNIQUE

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***Abstract* - In today's Scenario, voters must go to the polling booth to vote which takes time and the voting percentage is also low due to some voter's do not go to polling booth to cast the vote. Digital voting would be helpful as voter can vote online from their computer or laptop or smart phone which can save time of voter. It will be helpful to increase the voting percentage. Digital voting has admin login which will be handled by election commission and voter login which will be handled by voter. One voter can vote for a candidate only once per election. We use two levels of security: First level : the face of the voter is captured by web camera and sent to the database and verified with the face present in the database. The comparison of the two faces is done using Random forest algorithm Second level: A password (OTP) is used as the second level of security, after entering the one-time password generated to their mobile number it is verified and allow to vote.**

***Keywords*: Online voting system, Online election system, Face recognition, Random forest, OTP**

I . INTRODUCTION

The Election is the important for every country to make county security, peace and healthy environment. In election time different party candidate standing in front of the citizen to the particular area to won the election. Today voting is the manual process and slow process to count the each and every vote, like voting commission send the electronic ballot box will send the every city where the both is allocated and collect the vote every citizen. So our proposed system solves the existing system problem based on online voting system to take the voting online from any location. So it is a time consuming process and also requires a lot of resources to conduct voting process. In

this paper we have proposed online voting system to cast the vote using face recognition and OTP. The information about the OTP and Face is passed to the server unit for further verification. Then the server checks for the data from the database and compares that data which is already existing in database. If the data matches with the already stored information, the person is allowed to poll the vote. If not, a message is displayed on the screen and therefore the person is not allowed to poll the vote. For voting representatives are appointed by electorates. In current scenario voter needs to show his/her voter ID card to cast the vote on the booth. So this process is time consuming as the voter ID card needs to be get verified by the officials. Thus to speed up the voting process and avoid such type of problems, we have proposed the new system.

FIGURE 1. Polling System

RELATED WORK

The voting system without polling stations was discussed by P.Mccorry et al. He suggested that voting through the internet using blockchain can give good results if implemented correctly. They discussed some technical flaws in digital voting systems. The robustness of the system could not be controlled. The error of doubling users gets minimal by using end-to-end verification. These voting systems had low latency and did not secure the privacy of voters. This latency in the system is controlled by using a flexible consensus algorithm and smart contracts in the blockchain voting system being proposed.

Mahalakshmi Mabla naik,Dr.Preethi N.patil introduced a smart voting system through face recognition. The system use three degrees of confirmation which are user id confirmation, voter card number confirmation and face recognition confirmation. The benefits include enhanced security, accuracy, more easily accessible.

Ishwari Pawar, Shruti Rgade, Akshada Zaware proposed a system with multiple layers of verification to ensure the reliability of the device

which includes face verification and then OTP verification with validation data for voting.The benefits of the system include reduce the risk of fraudulent vote.

A proposed system by Nandan Gowda, Jayam Haresh Tharun improved the scalability, Maintenance and cost efficiently of the online voting system. It aims to build a smart voting system using face recognition technology that allows any voter in INDIA by going to their respective constituency from "ANYWHERE IN INDIA" to the nearest voting booth in the place of stay.

The voting system without polling booths was discussed by Chandra Keerthi Pothina, Alta Indu Reddy. It uses faces to unlock the system and does not require any physical presence to cast a vote as the traditional system does.

M.Pawlak proposed a system that does not require any operating entities .However, it could not secure a voter's identity and also require complex computing. The system was able to collect votes from users but due to complex computation, upon higher user rate the latency become an issue. The identities of the voter become vulnerable the system could not compute a large amount a data hence it has failed to be implemented at a large scale .Whereas in our proposed voting system the latency is managed by flexible use of consensus algorithm. The use of cryptographic hash in blockchain permits the risk of vulnerability of a voter's identity.

II.PROPOSED FRAMEWORK OF VOTING SYSTEM

The web application for voting process that is Online Voting System through Face Recognition and OTP that provides enhanced security, Cost effective and efficiency. The online voting system will manages the voter's details, Candidate details.The main feature of the project includes voters information and candidate information, voter can login and use his/her voting rights. The system can manage the information data very efficiently. The proposed system is more reliable,faster,accurate and easy to handle compared to existing manual system. It helps to computerize everything and reducing the errors as compare to manual voting system.

ARCHITECTURE DIAGRAM

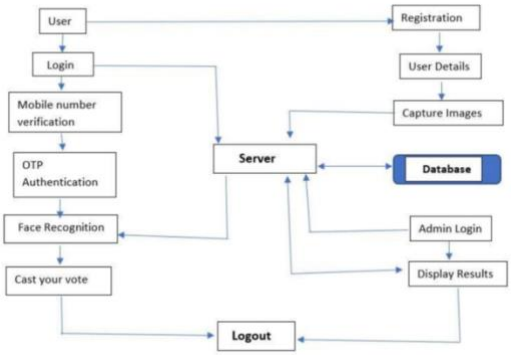

FIGURE 2. Architecture diagram of proposed system.

The above architecture provides two levels of security ie., the face recognition and OTP. The voting system is handled by the admin. The admin has the control of updating, adding and maintaining the list of the voters. The Comparison of the two faces is done using the machine learning algorithm ie., the random forest algorithm which can able to handle more number of large datasets i.e., the large number of voters involved in the voting process.

HARDWARE REQUIREMENTS

Processor: Pentium

RAM: 4GB

Hard Disk: 1TB

Speed: 1.1GHz

SOFTWARE REQUIREMENTS

Operating System: Windows

Scripting Language: PHP

Back-End: MYSQL

Front-End: HTML and CSS

Type: Web Application

Server: XAMP(Apache, MYSQL)

PHP version: PHP Version 7.0

WORKFLOW OF PROPOSED SYSTEM

The voting system architecture consists of two phases of authentication i.e., the face recognition and OTP level Security. The system consists of two types of login, Admin and voter login. The user will login into the system with their respective login name and password. The voter's

details are already stored in the database during the phase of registration. The user details and their images are stored in the database. The same details can be accessed through the server. Once the voter entered into the voting page , the voter will be asked to verify his face using web camera. Once the voter's face is verified, the first level of authentication is completed. Further, the voter is verified by entering the OTP send to their registered mobile number. At last, when all the levels of security are verified then the voter is allowed to vote. The admin has the control of making changes in the voter's database, adding, and updating the candidate list and displaying the results.

MODULES USED IN PROPOSED SYSTEM

1.IMPLEMENTATION OF ONLINE VOTING SYSTEM APPLICATION:

Online voting system has admin login which will be handled by the election commission and the voter login which will be handled by the voter. Voter can login through their Voter ID number and password after a successful registration.

Front end: HTML, CSS, JAVASCRIPT (DJANGO FRAMEWORK)

1.HTML: HTML is used to create and save web document. E.g., Notepad/Notepad++

2.CSS: (Cascading Style Sheets) Create attractive layout.

3.Bootstrap: responsive design mobile friendly site.

4.Javascript: It is a programming language, commonly use with web browsers.

Back end: PHP, MYSQL

1.PHP: Hypertext Pre-processor (PHP) is a technology that allows software developers to create dynamically generated web pages, in HTML, XML, or other document types, as per client request. PHP is open-source software.

2. MYSQL: MySQL is a database, widely used for accessing querying , updating and managing data in databases.

Software Requirement:

XAMPP Server

2. REGISTRATION OF THE CANDIDATES AND VOTER:

In this module, the individual candidate should register their party name and their party symbol. The information given by the individual candidate will be stored in the database by the electoral committee. The individual voter could cast their vote only if they had completed their registration. The voter must enter their name, Voter ID, Email ID, phone number and they need to upload the image of the individual. Once the image have been uploaded, the registration gets completed.

3.DETECTION AND RECOGNITION OF THE VOTER'S FACE :

In this module, the detection and recognition of the voter's face is done. During the time of voting, the face of the voter is detected using web camera and the face is compared with image that is already stored in the database. If the face stored in the database matches with the real time face , then the voter is allowed to vote.

4. OTP LEVEL SECURITY:

In this module, the voter gets a OTP send to their respective mobile number. The mobile number is provided by the voter at the time of the registration process. The individual voter should provide their valid number to cast their vote. Once the OTP is verified, the voter is allowed to vote. Once the voter has voted, the voting portal will be closed.

ALGORITHM

RANDOM FOREST (RF)

It is the process of Combining multiple classifier to solve the complex problem and improve the performance of the model.

Random forest is a classifier it contains a number of decision trees on various subsets of the given datasets and takes the average to improve the predictive accuracy of the dataset.

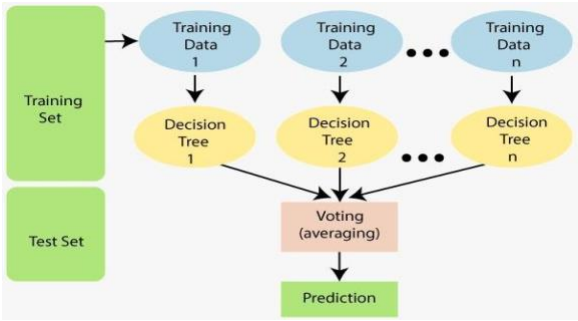

FIGURE 3. Diagram of Random forest Algorithm

Steps involved in the algorithm:

1.Data Collection: Collect the data that contains images of people's faces.

2. Labelling the data: Manually assigning labels to the images.

3. Feature extraction: Model learns to classify the images.

4.Train the Random Forest model: Use labelled data to train the random forest model.

5. Testing the model: Test the trained model on a set of test images.

6. Deployment: Once the model is trained, it can be deployed in the online voting system to recognize the faces of the voters.

CONCLUSION

At present our government is spending 583 crores for conducting a Lok Sabha election. This money is spent on issues such as security, electoral ballots etc. The average percentage of voting is less than 60%. Moreover, voting fraud can be easily done in the present system. Also, the percentage of literates coming to vote is very less. But with our system the money spent on elections can be reduced to less than 10 crores. Also, there is no chance of voter fraud and the money spent on security can be drastically decreased. People who have an internet connection at home with a web camera can vote without taking the strain to come to voting booths**.**

FUTURE WORK

In the future, we can make entire election system with a 'Live Result Update' feature.

This system that can also guarantee high level secrecy, security, and verifiability of a marked ballot transmitted over the Internet.

This Online Voting System can be advanced to be used by any organization in the world for conducting smooth and fair elections.

REFERENCE

[1] Anandaraj S., Anish R., and Devakumar P.V. (2015), 'Secured electronic voting machine using biometric', IEEE International conference on innovations in information, embedded and communication systems**.**

[2]Srivatsan sridharan (2013), 'Implementation of authenticated and secure online voting system', IEEE

International conference on computing, communication and networking technologies.

[3] Bishop M., and Frincke D. (2007), 'Achieving learning objectives through e-voting case studies', IEEE Security& Privacy, Vol. 5, No. 1, pp.53–56.

[4]Herb Deutsch (2005), 'Public opinion's influence on voting system technology', IEEE Standards Association.

[5]Hongyu Zhang., Qianzi You, and Junxing Zhang (2015), 'A lightweight electronic voting scheme based on blind signature and kerberos mechanisms', International conference on advanced networks and telecommuncations systems,pp.978-4799.

[6]Jambhulkar S.M., Jagdish B. Chakole, and Praful. R. Pardhi (2014), 'A Secure approach for web basedinternet voting system using multiple encryption', IEEE International conference on electronic systems, signal processing and computing technologies.

[7]Kausal malladi, Srivatsan sridharan, Jayprakash L.T. (2014), 'Architecting a large-scale ubiquitous e-voting solution for conducting government elections', IEEE International conference on advances in electronics, computers and communications.

[8]Kohno. T et al., (2004), 'Analysis of an electronic voting system', Proc. 2004IEEE symposium security and privacy, pp. 27–40.

[9]Mohammad Malkawi., Omar Al-Jarrah., Thaier S. Hayajneh., Munzer S. Ebaid, and Mohammed Khasawneh (2008), 'A biometric-secure eVoting system for election processes', IEEE symposium on mechatronics and its applicationsMay 27-29.

[10]Mona F.M.Mursi., Ghazy M.R.Assassa., Ahmed A. AbdelHafez, and Kareem M. AboSamra (2015), 'A secure and auditable cryptographicbased e-voting scheme', International conference on mathematics and computers in sciences and in industry, IEEE.